

Multimodal Visual Understand

Function Description

After the program runs, after waking up the voice module, ask questions about vision or describe the current scene. The program will take a photo of the current environment and upload it to the Alibaba Cloud platform. Then the vision model will parse the image with the question and finally provide an answer.

Open the terminal and enter the following command:

```
ros2 launch largemodel largemodel_control.launch.py
```

After waking up the module, give a vision-related command, reference the following example:

```
Please describe what do you see?
```

```
[asr-5] [INFO] [1775189938.401250893] [asr]:  
[asr-5] [INFO] [1775189938.075253188] [asr]: -  
[asr-5] [INFO] [1775189938.135225578] [asr]: -  
[asr-5] [INFO] [1775189932.399906328] [asr]: XunFei ASR Finish.  
[asr-5] [INFO] [1775189932.400602741] [asr]: Please describe what do you see?  
[asr-5] [INFO] [1775189932.401250893] [asr]: @okay, let me think for a moment...  
[model_service-3] [INFO] [1775189946.800359899] [model_service]: Decision making AI planning:seewhat()  
[model_service-3] [INFO] [1775189949.264987436] [model_service]: "json_str": {"response": "I'm looking around and taking in everything I see right now, let  
me show you what's in my view!", "action": ["seewhat()"]}  
[model_service-3] [INFO] [1775189949.265629644] [model_service]: "action": ['seewhat()'], "response": "I'm looking around and taking in everything I see righ  
t now, let me show you what's in my view!"  
[action_service-4] [INFO] [1775189949.292257439] [action_service]: "len(actions)": 1  
[action_service-4] [INFO] [1775189949.292705053] [action_service]: "actions": ['seewhat()']  
[model_service-3] [INFO] [1775189960.279968419] [model_service]: get an image...  
[model_service-3] [INFO] [1775189965.390117291] [model_service]: "json_str": {"response": "I see a neat setup here! There's a white keyboard with blue keys  
on the right, and a black mouse resting on a cute mousepad with a cartoon design. The mousepad has some pink markings on it too. It looks like a cozy worksp  
ace!", "action": []}  
[model_service-3] [INFO] [1775189965.390866518] [model_service]: "action": [], "response": "I see a neat setup here! There's a white keyboard with blue keys  
on the right, and a black mouse resting on a cute mousepad with a cartoon design. The mousepad has some pink markings on it too. It looks like a cozy worksp  
ace!"  
[action_service-4] [INFO] [1775189965.465900290] [action_service]: "len(actions)": 0  
[action_service-4] [INFO] [1775189965.466633314] [action_service]: "actions": []  
[action_service-4] [INFO] [1775189983.393653741] [action_service]: Published message: Robot feedback: Reply to user completed  
[model_service-3] [INFO] [1775189987.174892037] [model_service]: "json_str": {"response": "I've shared what I see, and it's such a nice setup! If you'd like  
me to focus on anything specific, just let me know!", "action": ["finish_dialogue()"]}  
[model_service-3] [INFO] [1775189987.175455226] [model_service]: "action": ['finish_dialogue()'], "response": "I've shared what I see, and it's such a nice s  
etup! If you'd like me to focus on anything specific, just let me know!"  
[action_service-4] [INFO] [1775189987.179648273] [action_service]: "len(actions)": 1  
[action_service-4] [INFO] [1775189987.180079647] [action_service]: "actions": ['finish_dialogue()']  
[action_service-4] [INFO] [1775189999.286058269] [action_service]: Published message: finish  
[model_service-3] [INFO] [1775189999.286488920] [model_service]: The current instruction cycle has ended
```

The path where the captured photo is saved is:

```
LargeModel_ws/install/largemodel/share/largemodel/resources_file/image.png
```



Core Code Analysis Seewhat

Please refer to the **Core Code Analysis Seewhat** content in tutorial **【AI Model-Text Version】 - 【2.Multimodal Visual Understand】** . The voice version and text version have the same action functions; only the input method for task commands is different.